B.Sc.Engg. CSE 3rd Semester

5th December 2022

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2021-2022**

DURATION: 3 HOURS **FULL MARKS: 150**

Math 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 6 (Six) questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

1. a) Consider the following matrix A: 10+2 (CO3) (PO1)

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

i. What would be projection matrix (P) of A to project any vector b onto the columnspace of A?

ii. Find P^3.

b) Table 1 represents the relationship between the timestamp (t) in hours and total unit of production (u) of an industry on any random day: 11+2 (CO2) (PO1)

Table 1: Table for Question 1.b)

Timestamp (t)	Total unit of production (u)
-2	1
0	2
2	4
4	5

Find a linear equation that fits these data by minimizing the error. Also, predict the unit of production (u) on the 8th hour of that particular day.

2. a) Consider the following matrix A: 13 (CO1) (PO1)

$$A = \begin{bmatrix} 3 & 4 & 6 \\ 0 & 1 & 0 \\ -1 & -2 & -2 \end{bmatrix}$$

Find a basis of R^3 consisting of eigenvectors of A.

b) Suppose A is a 3 by 3 matrix with eigenvalues 0, 1 and 2. Identify the following: 4×3 (CO3) (PO1)

i. the rank of A

ii. the determinant of A^TA

iii. the determinant of $A + I$

iv. the eigenvalues of $(A + I)^{-1}$

$\frac{1+\frac{1}{2}}{2}$

3. a) Consider the following sequence S: 7+7+6 (CO3) (PO1)

$$0, 1, \frac{1}{2}, \frac{3}{4}, \frac{5}{8}, \ldots\ldots\ldots\ldots$$

i. Find a matrix A that satisfies

$$\begin{bmatrix} F_{k+2} \\ F_{k+1} \end{bmatrix} = A \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}$$

where F_k denotes the k^{th} term of the above-mentioned sequence S.

ii. Diagonalize the matrix A so that you can easily produce the A^k for any number k. You do not need to multiply the decomposed elements to get a single matrix.

iii. Find the 100^{th} term of the sequence S.

b) Is it possible to choose all of the eigenvectors of a real symmetric matrix perpendicular to each other? Justify your answer. 5 (CO1) (PO1)

4. a) Consider the following matrix A: 8+7 (CO3) (PO1)

$$A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}$$

i. Find orthonormal vectors q_1, q_2 *and* q_3 such that q_1 *and* q_2 form a basis for the columnspace of A and q_3 remains in the left nullspace of A.

ii. Find the closest vector in the columnspace of A to $b = (1, 2, 7)$.

b) Find the determinants of the following matrices: 5+5 (CO3) (PO1)

$$C = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 0 & 5 & 7 \end{bmatrix} \qquad D = \begin{bmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{bmatrix}$$

5. a) Suppose A is a 3 by 3 matrix with eigenvalues 1, 0 and -1. The eigenvector matrix of A is the following matrix S: 10 (CO3) (PO1)

$$S = \begin{bmatrix} -1 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

If matrix B is $B = A^9 + I$, give a reason why the matrix B does have or doesn't have each of the following properties:

i. B is invertible

ii. B is symmetric

iii. $trace = B_{11} + B_{22} + B_{33} = 3$

b) Find the Singular Value Decomposition (SVD) of the following matrix A: 15 (CO3) (PO1)

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

6. a) Suppose matrix A is the following product: 12 (CO2) (PO1)

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 5 \\ 2 & 1 & 0 & 0 \\ 3 & 1 & 1 & 0 \end{bmatrix}$$

For what values of t (if any) are there solutions to $Ax = (1, 1, t)$?

b) Find the conditions on a and b that make the matrix A invertible, and find A^{-1} when it exists: 12 (CO1) (PO1)

$$A = \begin{bmatrix} a & b & b \\ a & a & b \\ a & a & a \end{bmatrix}$$

B.Sc. Engg. in CSE / SWE (3rd Semester) 27 August 2020

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2019-2020**

DURATION: 1 Hour **FULL MARKS: 45**

Math 4341: Linear Algebra

General Instructions

- Write your **Name, Student-ID** and **Course Code** on the top of the first page. Maintain a **serial number** on the Top-right corner of each page.
- Answer all the questions. Figures in the right margin indicate marks.
- Sit in proper position and maintain the environment as per the Guidelines.
- No examinee is allowed to scan the file unless 1 hour is finished.
- For any circumstances, follow the instructions of the invigilator.

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1 Suppose, a group of 4 students from IUT CSE'18 went for a tour. They bought a number of souvenirs from there. While packing, they found there were 4 different items. Suppose the items are named A, B, C & D. Everyone bought at least one item or more than that. Now, they tried to put this purchase information into a matrix (Purchase Matrix). So, the columns represent the number of a particular item they bought and the rows represent individual purchase information (number of items) of the students.

a. *It was found out that the items were arranged in the columns in this order- B->C->A->D. How do you put it in the correct order(alphabetical) in the matrix without directly manipulating the columns?* 5

b. The purchase details are as follows- 7

Items	Student 1	Student 2	Student 3	Student 4
A	2	0	1	1
B	0	1	0	2
C	1	0	1	0
D	0	2	0	5

Let's say, V_1=subspace generated by the first column of the purchase matrix
V_2=subspace generated by the third column of the purchase matrix
Find out V_1 & V_2(Draw). What is $V_1 \cap V_2$? Is it a vector subspace as well?

c. Mention the dimension of all four fundamental subspaces of the purchase matrix. What can be one good basis for the rowspace? 5

2 Time (in -th seconds) and number of errors done by a machine is given below:

Time (in -th second)	Number of errors
-2	1
0	2
2	4

a. Can you write these data points as a simple linear system? 2

b. Find a linear equation that fits these points by minimizing the error. 10

c. Can you predict the number of errors done by the machine on the 5th second? 3

3 The following system is given- 6

$$\left[\begin{array}{cc|c} 1 & 3 & 2 \\ 3 & h & k \end{array}\right]$$

Find out the values of ***h*** and ***k*** so that the system becomes-

a. Inconsistent. [No solution]
b. Consistent with infinitely many solutions.
c. Consistent with a unique solution.

4 Short Questions-

a. If a matrix has an eigenvalue with $\lambda=0$, what can you decide about the rank and columnspace of the matrix? Is it a matrix with full rank as column? Do the columns span the whole vectorspace? 4
b. If a matrix has two eigenvalues $\lambda_1=3$ and $\lambda_2=4$, 3
   i. Find out the determinant of the matrix.
   ii. Find out the sum of the diagonal elements of the matrix.

B.Sc. Engg. CSE 3rd Semester 22 December 2023 (Afternoon)

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SEMESTER FINAL EXAMINATION WINTER SEMESTER, 2022-2023

DURATION: 3 HOURS FULL MARKS: 150

Math 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

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1. a) Let $a, b \in \mathbb{R}$, and let 10+5 (CO1) (PO1)

$$A = \begin{bmatrix} 1 & 2 & 3 & a \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & b \end{bmatrix}$$

   i. What are the dimensions of the four subspaces associated with the matrix A? These values will depend on the values of a and b, and you should distinguish all cases.

   ii. For $a = b = 1$, give a basis for the columnspace of A. Is this also a basis for $\mathbb{R}^3$? Justify your answer.

   b) Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation satisfying: 10 (CO1) (PO1)

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -4 \\ 3 \end{bmatrix} \quad \text{and} \quad T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -10 \\ 8 \end{bmatrix}.$$

   Find $T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$.

2. a) If you convert matrix A (shown below) to row-reduced echelon form by the usual row elimination steps, you will achieve $R = rref(A)$ as: 20 (CO3) (PO1)

$$A = \begin{bmatrix} 1 & 2 & 1 & -7 \\ 2 & 4 & 1 & -5 \\ 1 & 2 & 2 & -16 \end{bmatrix} \Rightarrow R = \begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & -9 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

   i. What are the minimum and the maximum number of columns of A that form a dependent and independent set of vectors, respectively?

   ii. Give an orthonormal basis for the rowspace of A. Note that, it is not $C(A)$.

   iii. Given the vector $b = (2\ \ 5\ -9\ \ 3)^T$, compute a vector p that is closest to b in the rowspace $C(A^T)$.

   iv. In terms of your calculated p in Question 2.a)iii., what is the closest vector to b in the nullspace $N(A)$?

   b) What is the value of the following determinant for any values of a, b and c? 5 (CO1) (PO1)

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ b+c & c+a & a+b \end{vmatrix}$$

3. a) Imagine a plane $z = Cx + Dy + E$ that is closest (in the least squares sense) to these four measurements: 15 (CO2) (PO1)

$$\begin{aligned}&\text{At } x = 1; y = 0 \text{ measurement gives } z = 1\\ &\text{At } x = 1; y = 2 \text{ measurement gives } z = 3\\ &\text{At } x = 0; y = 1 \text{ measurement gives } z = 5\\ &\text{At } x = 0; y = 2 \text{ measurement gives } z = 0\end{aligned}$$

Show that this system $Ax = b$ has no solution. Find the best least squares solution $\hat{z} = (\hat{C}, \hat{D}, \hat{E})$.

b) A is an $m \times n$ matrix. Suppose $Ax = b$ has at least one solution for every b. 10 (CO1) (PO1)

i. What is the rank of A?

ii. Describe all vectors in the nullspace of A^T.

iii. Determine whether the equation $A^T y = c$ has (0 or 1), (1 or ∞), (0 or ∞), or (1) solution for every c.

4. a) Suppose matrix A has eigenvalues $\lambda_1 = 3, \lambda_2 = 1, \lambda_3 = 0$ with corresponding eigenvectors: 10 (CO2) (PO1)

$$x_1 = \begin{bmatrix}1\\1\\1\end{bmatrix} \quad x_2 = \begin{bmatrix}0\\1\\1\end{bmatrix} \quad x_3 = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

Find the matrix A.

b) Let V be a two-dimensional subspace in $\mathbb{R}^3$ consisting of vectors $[x \;\; y \;\; z]^T$ satisfying: 15 (CO1) (PO1)

$$x + 2y - 5z = 0$$

i. Find a 3×2 matrix A whose columnspace is V.

ii. Find an orthonormal basis for V.

5. a) Consider a sequence of numbers defined using the following terms: 5+10+5 (CO3) (PO1)

$$\begin{aligned}G_0 &= 0\\ G_1 &= 1/2\\ G_{k+2} &= (G_{k+1} + G_k)/2\end{aligned}$$

i. Set up a 2×2 matrix A to get from $\begin{bmatrix}G_{k+1}\\G_k\end{bmatrix}$ to $\begin{bmatrix}G_{k+2}\\G_{k+1}\end{bmatrix}$.

ii. Find an explicit formula for G_k.

iii. What is G_{1000}? What is the limit of G_k as $k \to \infty$?

b) Suppose a 3×3 matrix A has $row_1 + row_2 = row_3$. Explain why $Ax = [1 \;\; 0 \;\; 0]^T$ can not have a solution. 5 (CO1) (PO1)

6. a) Compute the Singular Value Decomposition $A = U\Sigma V^T$ for $A = \begin{bmatrix}0 & 1\\-1 & 1\\1 & 0\end{bmatrix}$. 15 (CO3) (PO1)

b) The matrix A has a varying $1 - x$ in the $(1,2)$ position: 10 (CO1) (PO1)

$$A = \begin{bmatrix}2 & 1-x & 0 & 0\\1 & 1 & 1 & 1\\1 & 1 & 2 & 4\\1 & 1 & 3 & 9\end{bmatrix}$$

Considering the properties of the determinant, show that $\det(A)$ is a linear function of x. For any x, compute $\det(A)$. For which value of x is the matrix singular?